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Current landscape of BRD4 inhibitors: Selective targeting and protein degradation for enhanced efficacy☆

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ABSTRACT

The bromodomain and extra-terminal domain (BET) family of proteins has been implicated as a crucial factor in the pathogenesis of numerous diseases by regulating gene expression and chromatin dynamics through its bromodomains, which bind to acetylated lysine residues. As a key member of the BET protein family, BRD4 has emerged as a potential therapeutic target for a wide spectrum of diseases, including cancer, inflammation, fibrosis, cardiovascular disorders, and viral infections. In recent years, pan-BET inhibitors have advanced in clinical studies, but their efficacy remains limited by toxicity, resistance, and pharmacokinetics. In contrast, BRD4-BD1 and BRD4-BD2 selective inhibitors show strong potential against viral, inflammatory, oncologic, and neurodegenerative diseases, with high efficacy, low toxicity, and favorable pharmacokinetics. Moreover, protein degradation strategies targeting BRD4 offer novel approaches to overcome drug resistance. This work reviews the current status of research on BRD4 inhibitors.

1. Introduction

Bromodomain-containing protein 4 (BRD4) is one of the most important members of the bromodomain. Compared with other BET family members (such as BRD2, BRD3, and BRDT) in terms of structure, one notable difference lies in its C-terminal domain (CTD), extra-terminal domain (BET) family. It plays a critical role in the initiation and progression of various diseases [1,2]. BRD4 specifically recognizes and binds to acetylated lysine residues (KAc) on histones and non-histone proteins through its two tandem N-terminal bromodomains (BD1 and BD2), (Fig. 1) thereby participating in chromatin dynamics and the transcriptional activation of proliferation-related genes [3,4]. as well as influencing physiological processes such as DNA damage repair [5–7]. thereby positioning BRD4 as a promising therapeutic target in cancer treatment [8], including Diffuse Large B-Cell Lymphoma (DLBCL)、 Acute Myeloid Leukemia (AML)、 Prostate Cancer(PCa)、 Triple-Negative Breast Cancer (TNBC). The pathogenesis of NUT midline carcinoma (NMC) critically depends on its involvement [9–13]. Aberrant activation of BRD4 is also implicated in the pathogenesis of inflammatory diseases, like Inflammatory Bowel Disease (IBD), periodontitis (PD), osteoarthritis (OA) and so on. It mainly acts in inflammation by controlling signaling pathways related to inflammation, such as NF-κB [14–17].

In recent years, the development of small-molecule inhibitors and protein degraders targeting BRD4 has made significant progress. Several pan-BET inhibitors have advanced into clinical trials; however, their application is limited by challenges such as dose-limiting toxicity (DLT). To address these issues, Researchers are increasingly focusing on creating inhibitors that selectively target BRD4, particularly those with selectivity for either BD1 or BD2 domains [18,19]. Recently, computational approaches, including molecular docking, molecular dynamics simulations, and free energy calculations, have played an important role in BRD4 inhibitor research. These methods have identified high-affinity quinoline derivatives (e.g., compound 4 g) with superior binding properties compared to standard inhibitors, providing a theoretical basis and potential leads for novel BRD4 inhibitor development [20,21]. Moreover, protein degradation approaches—such as proteolysis targeting chimeras (PROTACs) and molecular glues—have led to the development of BRD4 degraders, offering novel strategies to address drug resistance [22]. A comprehensive analysis of BRD4's biological roles, ongoing

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clinical studies, resistance mechanisms, and emerging inhibitors is presented in this review, offering updated insights into BRD4-targeted therapy.

2. Clinical landscape and key findings

Although notable progress has been achieved in the development of pan-BET inhibitors in recent years, most clinical trials are still in their early phases [23]. Several inhibitors have demonstrated notable toxicity during treatment. A major concern is dose-limiting toxicity (DLT), particularly adverse events such as thrombocytopenia, diarrhea, and hyperbilirubinemia, with thrombocytopenia being the most commonly observed adverse event [24,25]. In addition, potential immune deficiencies have also been reported as adverse effects [1,26]. Recent studies suggest that the toxicity associated with pan-BET inhibitors may stem from their non-selective inhibition of all BET family members, which leads to widespread dysregulation of gene expression [27]. In terms of antitumor activity, the clinical efficacy of some inhibitors has fallen short of their promising preclinical results. Poor target selectivity, the emergence of resistance in solid tumors, and suboptimal pharmacokinetic properties are now widely recognized as key factors limiting the development of BRD4 inhibitors. Several clinical candidates-including BAY 1238097 (NCT02369029), OTX-015 (NCT02296476 and NCT02698176), and I-BET151 (NCT02630251)—have been discontinued due to safety concerns in trials involving various lymphomas and other solid tumors [28,29].

Most BRD4 inhibitors in current trials (Table 1) are in phase I or I/II for solid tumors, hematologic malignancies, and rare cancers. Nonselective pan-BET agents such as OTX-015 and CPI-0610 have been halted due to safety and efficacy limitations. The BD2-biased inhibitor apabetalone (Resverlogix) showed 24 weeks of safe dosing in a phase III cardiovascular trial, supporting domain selectivity as a means to reduce systemic toxicity. BMS-986158 and FT-1101 raised the maximum tolerated dose by 1.5 to 2-fold in phase I hematologic studies using MYC-high or NUT-fusion enrichment. EP31670 and ZEN003694 have focused biomarker-driven use on Castration-Resistant Prostate Cancer(CRPC) and ovarian cancer, achieving over 40 % disease control without severe thrombocytopenia. These results indicate a shift from broad pan-BET inhibition to precision BRD4 targeting with BD1/BD2 selectivity or BRD4-specific degradation and strict molecular stratification.

3. Mechanisms of drug resistance

BRD4's BD1 and BD2 domains are each composed of four antiparallel α-helices (αZ to αC) of varying lengths. Differences in amino acid residues, polarity, and hydrophobicity within the binding pockets contribute to the poor selectivity of BET inhibitors [30]. Mechanistically, resistance mechanisms include BRD4 phosphorylation changes, altered protein stability, and pathway compensation. Notably, WNT/β-catenin and Hedgehog signaling can independently sustain MYC expression, leading to BRD4 inhibitor resistance [28,29,31]. In phosphorylation regulation, CK2-mediated BRD4 hyperphosphorylation drives drug resistance by activating transcription factors such as c-MYC and RAC1; in NUT midline carcinoma, CDK9 promotes BRD4 hyperphosphorylation, triggering oncogenic transformation; in triple-negative breast cancer, loss of PP2A-mediated dephosphorylation leads to sustained BRD4 activation; moreover, a phosphorylation switch modulates BRD4 chromatin-binding sites, influencing the recruitment of factors such as p53 [32–35]. In JQ1-resistant AML cells, resistance to BRD4 inhibitors has been partly attributed to adaptive apoptotic signaling mediated through AMPK/ULK1-dependent autophagy [36]. Protein stability regulation also contributes to intrinsic resistance. BRD4 protein turnover is directly regulated by the E3 ubiquitin ligase speckle-type POZ protein (SPOP) and the deubiquitinating enzyme DUB3, a mechanism that has been validated in CRPC models [37–40]. Furthermore, mutations in autophagy-related genes have been shown to activate downstream signaling cascades that attenuate the antitumor efficacy of BRD4 inhibitors [41].

4. Development of next-generation inhibitors

Compared to pan-BET inhibitors, BRD4-selective inhibitors exhibit

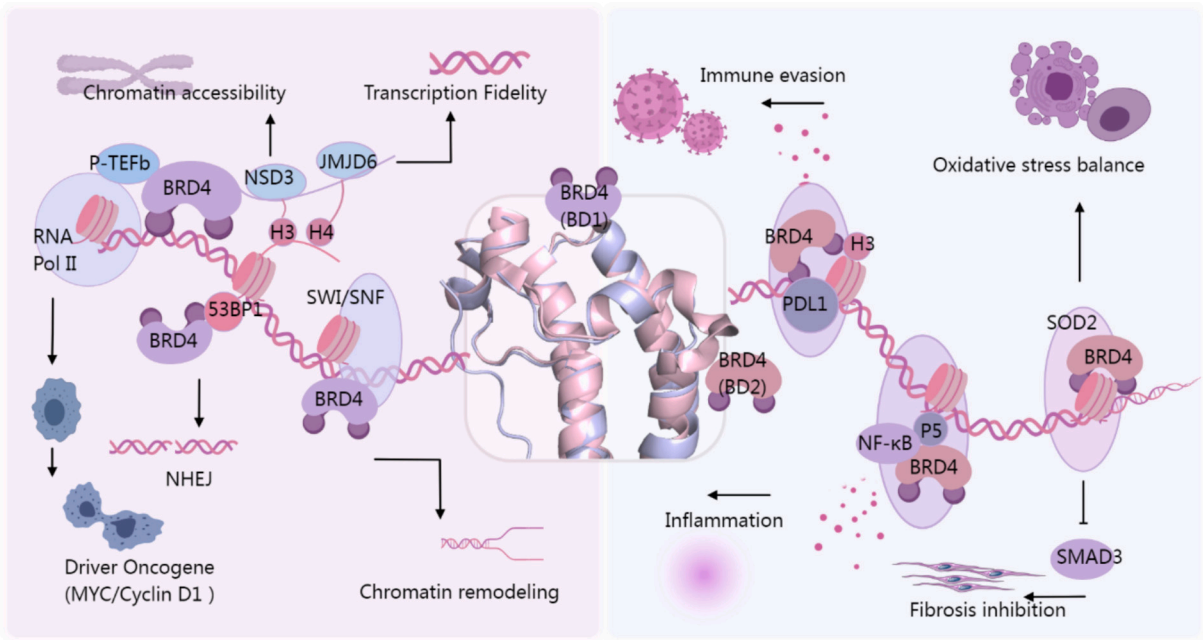
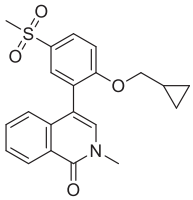
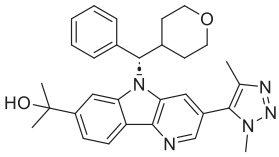
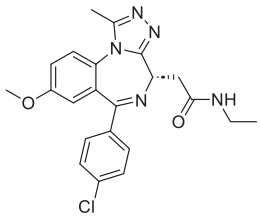
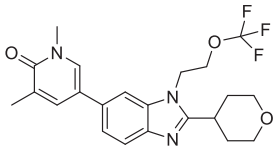
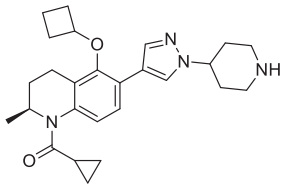
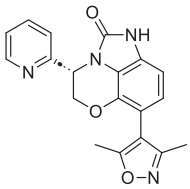
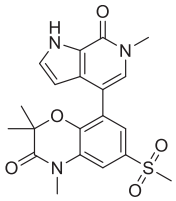
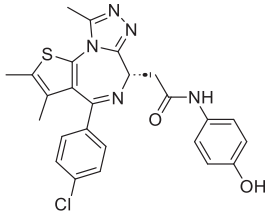


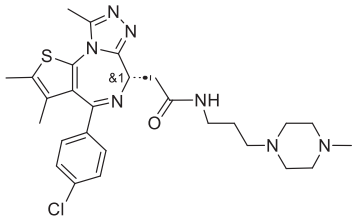
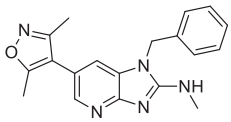
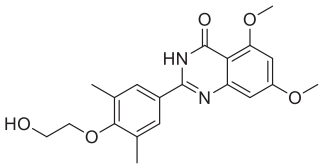
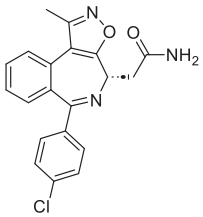
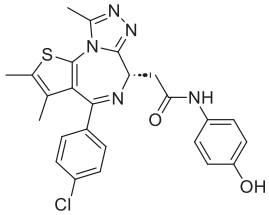
Fig. 1. The functional mechanism of BRD4:BRD4 contains two bromodomains (BD1 and BD2) that bind acetylated lysines on histone and non-histone proteins. Left: BD1 (purple) is a transcription hub: it recruits SWI/SNF to open chromatin, anchors P-TEFb to release Pol II pausing, ensures transcription fidelity via JMJD6, and regulates NHEJ through 53BP1. Right: BD2 (pink) functions as an acetylation amplifier. BD2 amplifies the expression of SDO2 (oxidative stress), NF-κB target genes (inflammation), PD-L1 (immune evasion), and SMAD3 (fibrosis). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
BRD4 inhibitors in Cancer-Related Clinical Trials.

Drug (Alias)	Chemical Class	NCT Number	Phases	Conditions	Sponsor	Study Status
BMS-986378		NCT03936465	I	Solid tumors, lymphoma, pediatric brain tumors	Dana-Farber Cancer Institute	COMPLETED
BMS-986158		NCT03936465	I	Solid tumors, lymphoma, pediatric brain tumors	Dana-Farber Cancer Institute	COMPLETED
		NCT02419417	I , II	Advanced Tumors	Bristol-Myers Squibb	COMPLETED
GSK525762C (Molibresib)		NCT03925428	I	Advanced lymphoma, solid tumors, refractory cancers, and pancreatic cancer	National Cancer Institute (NCI)	WITHDRAWN
		NCT01587703	I	Carcinoma, Midline		COMPLETED
		NCT01943851	II	Neoplasms		COMPLETED
GSK525762		NCT03266159	II	Solid Tumors	GlaxoSmithKline	WITHDRAWN
EP31670		NCT05488548	I	Castrate-resistant prostate cancer, NUT carcinoma, chronic myelomonocytic leukemia, Myelofibrosis (MF)	Epigenetix, Inc.	RECRUITING
FT-1101		NCT02543879	I	Myelodysplastic Syndrome(MDS), AML, non-Hodgkin lymphoma(NHL)	Forma Therapeutics, Inc.	COMPLETED
INCB054329		NCT02431260	I , II	Solid Tumors and Hematologic Malignancy	Incyte Corporation	TERMINATED
INCB057643		NCT02711137	I , II	Solid Tumors		TERMINATED
		NCT04279847	I	MF (various types), MDS, Myeloproliferative Neoplasm (MPN), Essential Thrombocythemia (ET)	Incyte Corporation	RECRUITING
		NCT06619522	II	MF	M.D. Anderson Cancer Center	WITHDRAWN
OTX015		NCT02303782	I , II	AML	Oncoethix GmbH	WITHDRAWN

(*continued on next page*)

Table 1 (*continued*)

Drug (Alias)	Chemical Class	NCT Number	Phases	Conditions	Sponsor	Study Status
RO6870810		NCT03068351	I	Multiple Myeloma	Hoffmann-La Roche	COMPLETED
ZEN003694		NCT02711956	I , II	Metastatic Castration-Resistant Prostate Cancer (mCRPC)	Zenith Epigenetics	COMPLETED
		NCT05071937	II	Ovarian /Peritoneal /Fallopian Tube Cancer	Alexander B. Olawaiye, MD	RECRUITING
		NCT04840589	I	Metastatic/recurrent/refractory ovarian carcinoma	National Cancer Institute (NCI)	RECRUITING
		NCT05803382	I	Metastatic colorectal carcinoma, unresectable colorectal cancer		
		NCT06102902	I	Metastatic/recurrent/refractory colorectal adenocarcinoma		
		NCT05053971	I , II	Advanced lymphoma, pancreatic cancer, refractory cancers		
		NCT05111561	I	Advanced/metastatic/refractory solid tumors, TNBC		
		NCT05372640	I	Breast Carcinoma , Malignant Solid Neoplasm		
		NCT05950464	I	Endometrial carcinoma, ovarian carcinoma, platinum-resistant ovarian carcinoma		
Apabetalone		NCT02586155	III	Diabetes Mellitus-Type 2 , Coronary Artery Disease	Resverlogix Corp	COMPLETED
CPI-0610 (Pelabresib)		NCT01949883	I	Lymphoma	Constellation Pharmaceuticals	COMPLETED
		NCT02157636	I	Multiple Myeloma		
		NCT02158858	I , II	preleukemia, precancerous conditions, neoplasms, MF, leukemia, acute myelocytic leukemia, MPN, MDS, primary myelofibrosis, myeloproliferative disorders, bone marrow disease, hematological disease,		
		NCT04603495	III	MF (various types)		
		NCT06401356	III	Hematologic Malignancy , Solid Tumor , Advanced Malignancies		RECRUITING
		NCT02986919	II	Peripheral Nerve Tumors	University of Texas Southwestern Medical Center	WITHDRAWN
MK-8628 (Birabresib)		NCT02698189	I	AML (De and Secondary), DLBCL	Merck Sharp & Dohme LLC	TERMINATED
		NCT02698176	I	NMC, TNBC, Non-Small Cell Lung Cancer (NSCLC), CRPC		
		NCT02259114	I	NMC, TNBC, NSCLC, Anaplastic Lymphoma Kinase(rearranged/ fusion), KRAS mutation, CRPC, pancreatic ductal adenocarcinoma	Oncoethix GmbH, a subsidiary of Merck & Co., Inc. (Rahway, New Jersey, USA)	COMPLETED
		NCT02296476	II	Glioblastoma Multiforme		TERMINATED

Sourced from https://clinicaltrials.gov/ (JUNE 2025).

improved target specificity toward BRD4 while exerting minimal inhibitory effects on other BET family members. BRD4's BD1 and BD2 domains have distinct functions and therapeutic relevance. BD1 is critical for chromatin binding, and its selective inhibition can mimic pan-BET effects with lower toxicity, showing promise in antiviral and anti-inflammatory therapy. BD2 mediates BET protein recruitment and induces interferon-related genes; its inhibition is beneficial in autoimmune and inflammatory diseases and may prevent radiation-induced fibrosis [27,42,43]. Consequently, considerable efforts have focused on developing domain-selective inhibitors. It is now generally recognized that achieving selective inhibition of either BD1 or BD2 is more attainable than targeting a single BET family member selectively [27]. Beyond domain-specific inhibitors, novel strategies—including dual-target inhibitors, proteolysis-targeting chimeras (PROTACs), and protein–protein interaction (PPI) inhibitors—have shown strong antitumor efficacy in preclinical models. These strategies represent promising directions for future drug discovery and offer alternative therapeutic avenues for overcoming the limitations of conventional BET inhibition [25].

4.1. Non-selective BRD4 inhibitors

Non-selective BRD4 inhibitors(Fig. 2), which include multi-target and dual-target agents, are being developed with increasing diversity. Researchers utilize diverse methods like virtual screening and structure–activity relationship (SAR) analyses to discover a broad spectrum of compounds (1–3). These molecules generally lack clear selectivity for the BD1 or BD2 domains but exhibit potent pan-BET inhibitory activity ($IC_{50} < 100$ nM), attracting significant attention due to their multi-target effects [44–46]. Multiple organizations have strategically developed patent portfolios encompassing diverse chemical scaffolds, including JQ1 derivatives, benzimidazoles, pyrrolopyridines, and sulfonamides. Notable examples include benzothiazole derivatives disclosed in EP 3936507 (4), α,β-unsaturated carbonyl covalent inhibitors reported in US 20210206779 (5), and diamino-pyrimidine sulfonyl derivatives described in US 10730860 (6), all of which demonstrate significant BRD4 inhibitory activity [47–49]. Of particular interest is the AR/BET dual-target inhibitor developed under WO2020063968 (7), which simultaneously suppresses the transcriptional activity of prostate-specific antigen (PSA) and FK506 binding protein 5 (FKBP5), showcasing innovative and potentially breakthrough therapeutic properties [50]. Non-selective BRD4 inhibitors cause dose-limiting toxicities such as thrombocytopenia and diarrhea due to their non-selectivity. Additionally, issues like solid tumor resistance (through MYC bypass and BRD4 phosphorylation escape) and pharmacokinetic shortcomings have forced the termination of most clinical trials.

4.2. BRD4-selective inhibitors

Apart from BRD4, bromodomain-containing protein 2 (BRD2), bromodomain-containing protein 3 (BRD3), and bromodomain-containing protein T (BRDT) are also members of the BET family [51,52]. BRDT, which is specific to germ cells, plays a key role in spermatogenesis and oogenesis through interactions with multiple co-factors, while BRD2 primarily regulates E2F-dependent gene expression involved in cell differentiation and proliferation. BRD3 is implicated in the transcriptional control of genes essential for embryonic stem cell development [53–55]. Due to the complex physiological functions of BET proteins, enhancing the selectivity of BRD4 inhibitors is crucial to minimize off-target toxicity and unwanted effects [30]. In recent years, a range of BRD4-selective inhibitors based on diverse chemical scaffolds (8–16, Fig. 3) have been reported. Among them, ZL0454 (8) and ZL0420 (9) were among the first identified BRD4-selective inhibitors [56]. Numerous inhibitors have since been developed for breast and ovarian cancers, exhibiting encouraging results in vitro and in vivo (10–16) [57–62]. A peptide inhibitor named PiET, targeting the extra-terminal (ET) domain of BRD4, has also been developed, offering a new approach for specific BRD4 inhibition [63].

4.3. BRD4-BD1-selective inhibitors

The BD1 domain of BRD4, essential for chromatin binding, has been selectively inhibited to achieve pan-BET-level efficacy while minimizing toxicity [42,64]. To date, a variety of BD1-selective inhibitors (17–23, Fig. 4) have been identified [65–67]. These inhibitors have demonstrated vigorous activity across multiple in vitro and ex vivo models. For instance, ZL0580 (19) has been shown to silence human immunodeficiency virus (HIV) expression epigenetically [65]. Other compounds, such as LT052 (20), ZL0513 (21), ZL0516 (22), and ZL0590 (23), exhibit enhanced selectivity toward BRD4-BD1 and show favorable characteristics, including improved metabolic stability, anti-inflammatory effects, and favorable pharmacokinetics and oral bioavailability [68–70].

Several additional BD1-selective inhibitors (24–28, Fig. 4) have also emerged recently, demonstrating potential in anti-inflammatory and antiproliferative applications, particularly in diseases such as AML [64,71–73]. Compounds including DW-71177 (29), GSK778 (30), GSK789 (31), compound (32), XL-126 (33), and DDO-8958 (34) have all exhibited high BD1 selectivity and effectively suppressed the growth of cancer cell lines such as breast and pancreatic cancers [74–77]. Notably, compound 32 can cross the blood–brain barrier and demonstrates strong anti-inflammatory effects in microglial cells, indicating its potential for treating multiple sclerosis (MS) [75]. Some compounds have also demonstrated synergistic antitumor effects when used in combination therapies; for example, DDO-8958 (34) combined with the glycolysis inhibitor 2-deoxy-D-glucose (2-DG) [77].

Internationally, several organizations have filed patents for BD1-selective compounds, many of which are based on scaffolds such as dihydroquinolinones, benzodihydropyridinones, azafluorenones, and purine derivatives [78–81]. Notable examples include compound 35 from patent WO 2023196455, which highlights the importance of amide and phenyl-pyridine groups in enhancing potency and selectivity [82]; compound 36 from CN 113121528, which features a fused KAc-

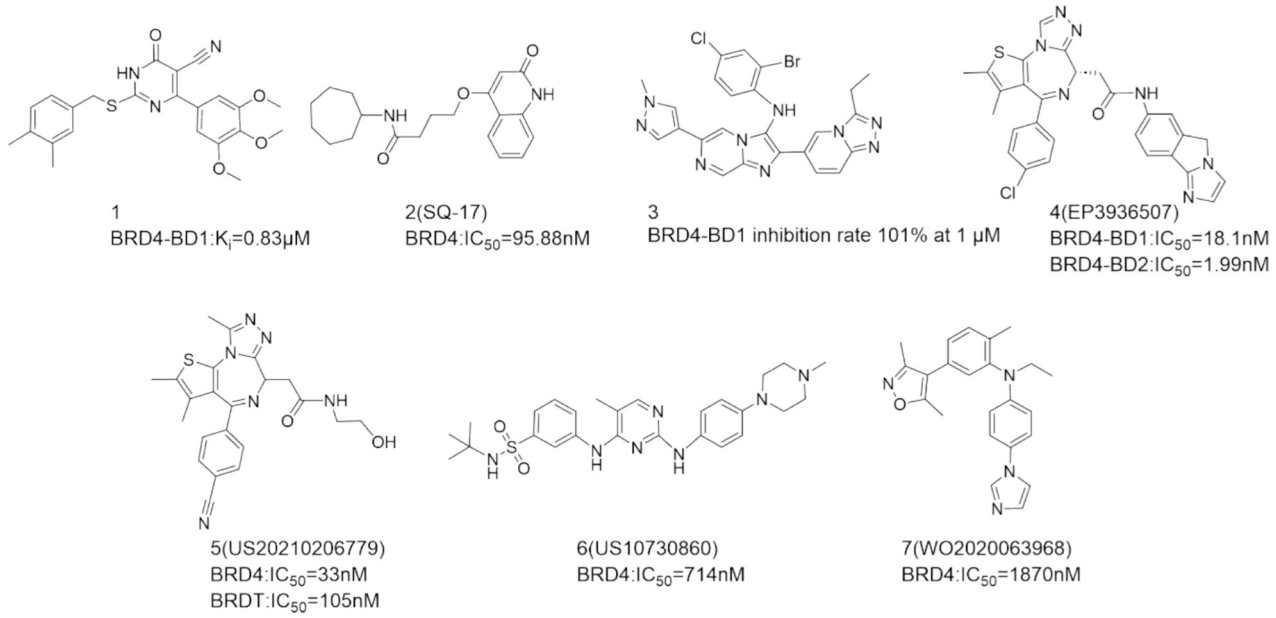
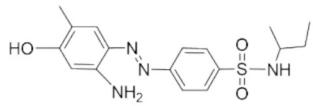
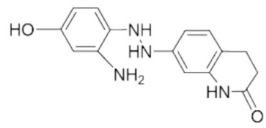


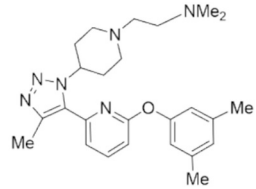
Fig. 2. Chemical structures of BRD4 Non-selective inhibitors 1–7.



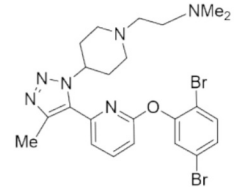
8(ZL0454)
BRD4−BD1:IC_{50}=49nM
BRD4−BD2:IC_{50}=32nM
BRD2−BD1:IC_{50}=772nM
BRD2−BD2:IC_{50}=1836nM
BRD3−BD1:IC_{50}=2493nM
BRD3−BD2:IC_{50}=2241nM
BRDT−BD1:IC_{50}=3292nM
BRDT−BD2:IC_{50}=3082nM



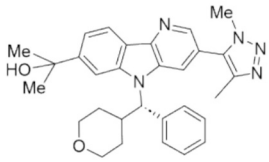
9(ZL0420)
BRD4−BD1:IC_{50}=27nM
BRD4−BD2:IC_{50}=32nM
BRD2−BD1:IC_{50}=803nM
BRD2−BD2:IC_{50}=1736nM
BRD3−BD1:IC_{50}=2275nM
BRD3−BD2:IC_{50}=2193nM
BRDT−BD1:IC_{50}=3183nM
BRDT−BD2:IC_{50}=2781nM



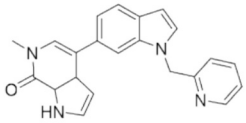
10(DW34)
BRD4−BD1:IC_{50}=0.137μM
BRD4−BD2:IC_{50}=1.02μM
BRD2−BD1:K_d=22nM
BRD2−BD2:K_d=200nM
BRD3−BD1:K_d=20nM
BRD3−BD2:K_d=310nM
BRDT−BD1:K_d=15nM
BRDT−BD2:K_d=250nM



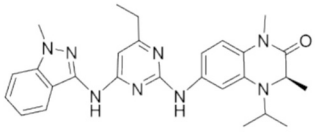
11
BRD4−BD1:IC_{50}=1.02μM
BRD4−BD2:IC_{50}=0.856μM



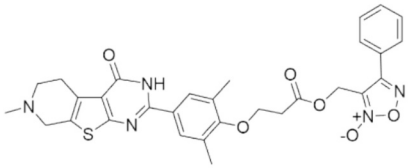
12(BMS-986158)
BRD4:IC_{50}=1.1nM



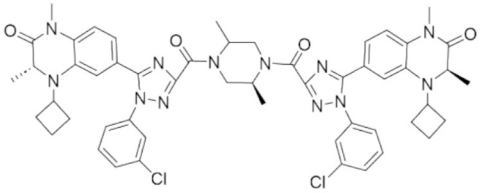
13(ZLD2218)
BRD4:IC_{50}=0.107μM
BRD2:IC_{50}=0.718μM
BRD3:IC_{50}=0.502μM
BRDT:IC_{50}>1μM
BRD1:IC_{50}>1μM



14
BRD4−BD1:IC_{50}=16.1nM
BRD4−BD2:IC_{50}=142.18nM
BRD2−BD1:IC_{50}=29.35nM
BRD2−BD2:IC_{50}=302.35nM



15(11a)
BRD4−BD1:IC_{50}=0.82μM
BRD4−BD2:IC_{50}=1.94μM
BRD2−BD1:IC_{50}=15.91μM
BRD2−BD2:IC_{50}>20μM
BRD3−BD1:IC_{50}>20μM
BRD3−BD2:IC_{50}>20μM
BRDT−BD1:IC_{50}>20μM



16
BRD4:IC_{50}=NA

Fig. 3. Chemical structures of BRD4 selective inhibitors 8–16.

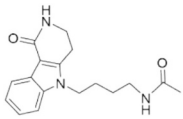
mimicking moiety targeting multiple tumor-related signaling pathways [83]; OTX015 (37) from WO 2020001152, which demonstrates picomolar BD1 inhibition; and a deuterated derivative (38) with BD1-selective inhibitory activity disclosed in WO 2022156757 [84,85].

4.4. BRD4-BD2-selective inhibitors

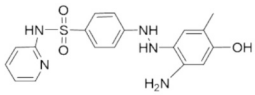
BRD4's BD2 domain drives BET protein recruitment to chromatin and activates interferon gene expression. Selective BD2 inhibition offers therapeutic potential for autoimmune and inflammatory diseases and may help prevent radiation-induced fibrosis [27,42]. Representative compounds are illustrated in (39–58, Fig. 5). Among them, GSK340 (39), a tetrahydroquinoline derivative, exhibits superior pharmacokinetic properties and druggability compared to its structural analogs [86]. The structurally optimized BY27 (40) suppresses tumor growth by modulating the c-MYC/p21 signaling axis and demonstrates low toxicity with improved selectivity [87]. ABBV-744 (41), a clinical candidate, is currently undergoing phase I clinical trials. Although iBET-BD2 (42) shows limited efficacy in suppressing tumor cell proliferation, it has been shown to regulate T helper 17 (Th17) cell differentiation, offering novel therapeutic potential in inflammatory diseases [88]. Tetrahydroquinoline analogs SJ018 (45) and SJ432 (46) have demonstrated antitumor potential in MYC-driven pediatric malignancies [89].

Among recent advances, the development of the furanopyridinone-based compound XY221 (47) [90]. and the isoflavone-derived compound 48 [18]. represents a promising direction for precise therapeutic strategies targeting immune-related disorders GSK097 (49), a pyridylamide-based inhibitor, exhibits excellent BD2 selectivity along with favorable liver microsome stability ($t_{1/2}$ > 120 min) and solubility (>200 μM at pH 7.4), meeting key criteria for clinical development [91,92]. Compounds GSK452 (50), GSK737 (51), and GSK217 (52), developed based on a 1,2,3-triazole core, demonstrate low lipophilicity, high solubility, and robust pharmacokinetic profiles [93].

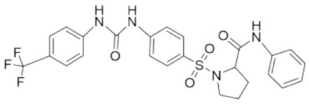
Other reported BD2-selective BET inhibitors include compound 53



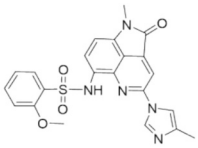
17(Olinoe)
BRD4−BD1:K_d=3.4μM
BRD4−BD2:K_d≥75μM



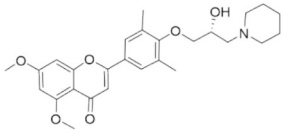
18(MS436)
BRD4−BD1:K_i≤0.085μM
BRD4−BD2:K_i≤0.34μM



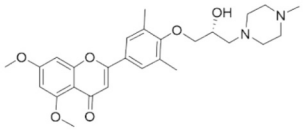
19(ZL0580)
BRD4−BD1:IC_{50}=163nM
BRD4−BD2:IC_{50}=1071nM



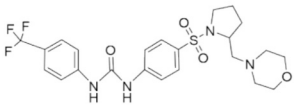
20(LT052)
BRD4−BD1:IC_{50}=87.7nM
BRD4−BD2:IC_{50}=12130nM



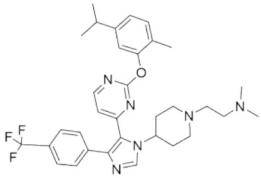
21(ZL0513)
BRD4−BD1:IC_{50}=67nM
BRD4−BD2:IC_{50}=684nM



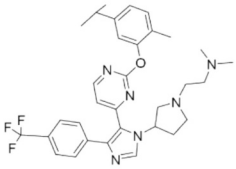
22(ZL0516)
BRD4−BD1:IC_{50}=84nM
BRD4−BD2:IC_{50}=718nM



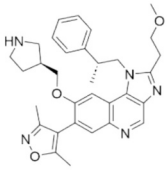
23(ZL0590)
BRD4−BD1:IC_{50}=90nM
BRD4−BD2:IC_{50}=1093nM
BRD2−BD1:IC_{50}=835nM
BRD2−BD2:IC_{50}=1308nM
BRD3−BD1:IC_{50}=2183nM
BRD3−BD2:IC_{50}=1996nM
BRDT−BD1:IC_{50}=2416nM
BRDT−BD2:IC_{50}=1902nM



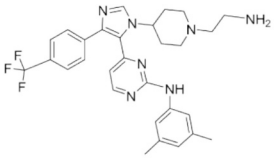
24
BRD4−BD1:IC_{50}<0.092μM
BRD4−BD2:IC_{50}>50μM



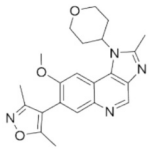
25
BRD4−BD1:IC_{50}<0.092μM
BRD4−BD2:IC_{50}>50μM



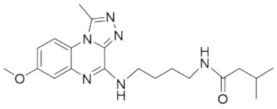
26
BRD4−BD1:pIC_{50}=5.2
BRD4−BD2:pIC_{50}=7.4



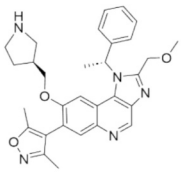
27
BRD4−BD1:IC_{50}=0.13nM
BRD4−BD2:IC_{50}>100μM



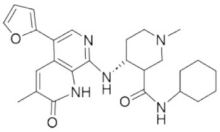
28
BRD4−BD1:IC_{50}=1.9nM
BRD4−BD2:IC_{50}=3.4nM



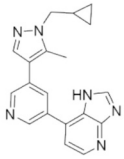
29(DW-71177)
BRD4−BD1:IC_{50}=20nM
BRD4−BD2:IC_{50}=293nM



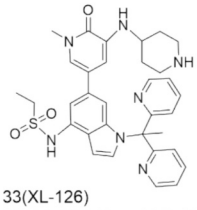
30(GSK778)
~130-flod selectivity for
BD1 over BD2



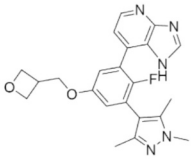
31(GSK789)
BRD4−BD1:IC_{50}=20nM
BRD4−BD2:IC_{50}=30μnM



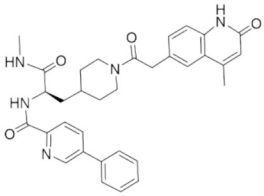
32
BRD4−BD1:IC_{50}=176nM
BRD4−BD2:IC_{50}=4837nM
BRD2−BD1:IC_{50}=497.9nM
BRD2−BD2:IC_{50}=2092nM
BRD3−BD1:IC_{50}=183nM
BRD3−BD2:IC_{50}=1940nM
BRDT−BD1:IC_{50}=4746nM
BRDT−BD2:IC_{50}=3043nM



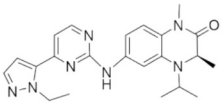
33(XL-126)
BRD4−BD1:IC_{50}=11.9nM
BRD4−BD2:IC_{50}=3110nM



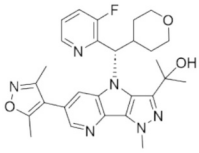
34(DDO-8958)
BRD4−BD1:IC_{50}=19nM
BRD4−BD2:IC_{50}=1220nM
BRD2−BD1:IC_{50}=25nM
BRD2−BD2:IC_{50}=1278nM
BRD3−BD1:IC_{50}=19nM
BRD3−BD2:IC_{50}=1220nM
BRDT−BD1:IC_{50}=1330nM
BRDT−BD2:IC_{50}=1825nM



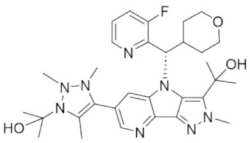
35(WO 2023196455)
BRD4−BD1:IC_{50}=4.4nM
BRD4−BD2:IC_{50}=2640nM



36(CN 113121528)
BRD4−BD1:IC_{50}<50nM



37(WO 2020001152)
BRD4−BD1:IC_{50}=0.35nM



38(WO 2022156757)
BRD4−BD1:IC_{50}=0.18nM

Fig. 4. Chemical structures of BRD4-BD1 selective inhibitors 18–38.

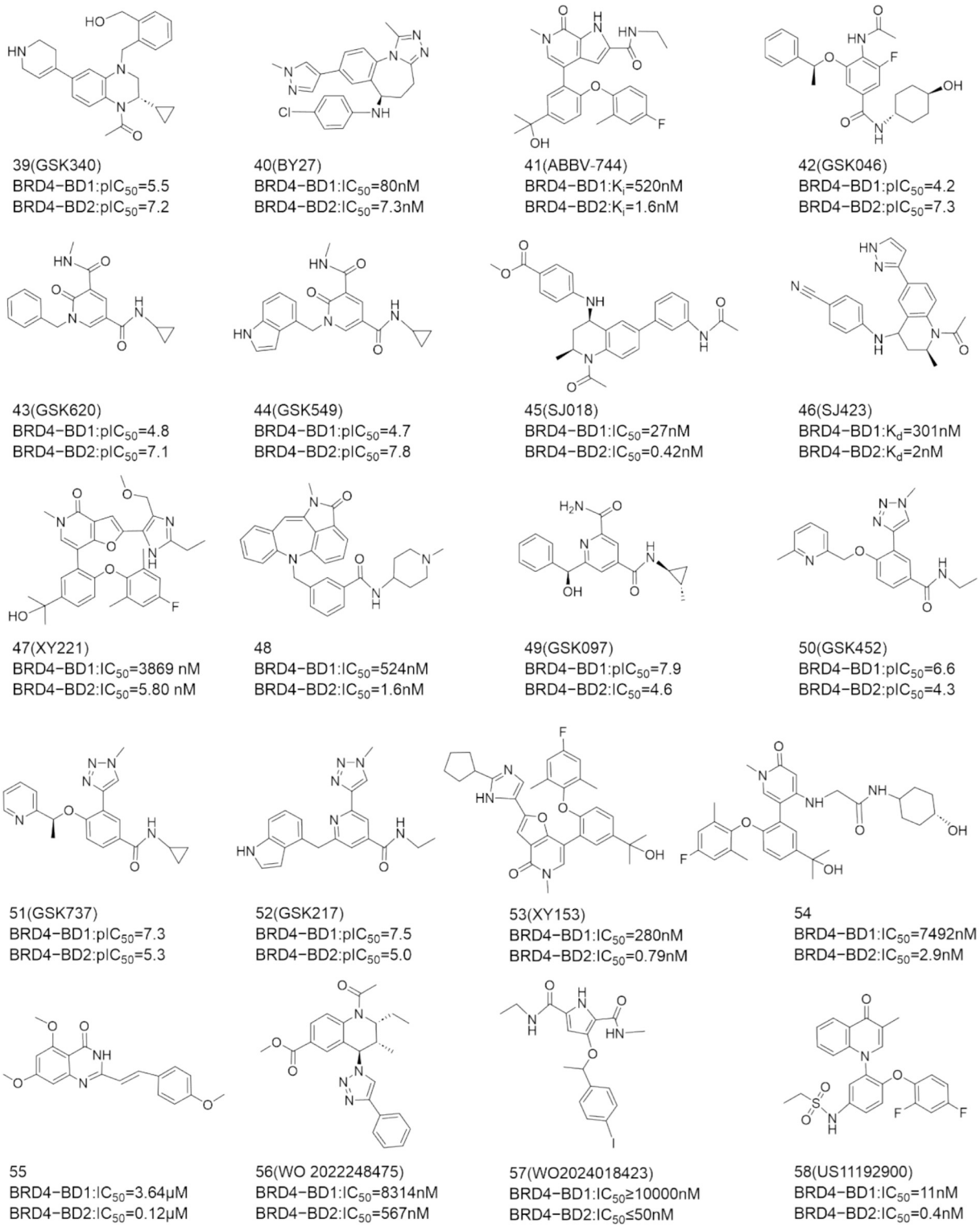


Fig. 5. Chemical structures of BRD4-BD2 selective inhibitors 39–58.

(XY153) and compound 54, both of which have demonstrated promising antitumor effectiveness and potential for treating AML. Compound 54 has demonstrated good cardiovascular safety in both cell-based and animal models [94,95]. Inspired by penipanoid C, Chen et al. synthesized compound 55, a selective BRD4-BD2 inhibitor exhibiting anti-inflammatory activity [96].

Multiple institutions, including Nuvation Bio Inc., Centre Hospitalier Universitaire, and Tay Therapeutics Ltd., have disclosed various BD2-selective inhibitors in their patents. These compounds predominantly feature chemical scaffolds such as dihydropyrrolopyridinones, pyrrolopyridinones, and tetrahydroisoquinolinone scaffolds. For example, patent WO 2022248475 describes tetrahydroquinoline (THQ)-based BRD4 inhibitors (56) [97], while WO 2024018423 presents BRD4 inhibitors incorporating pyrrole or imidazole as core structures (57) [98]. US11192900 reports pyrrolone derivatives (58) that exhibit enhanced kinetic stability and target selectivity [99].

4.5. Dual/multi-target inhibitors

In recent years, multi-target strategies have garnered significant attention in the development of anticancer therapeutics, particularly those that adopt polypharmacological approaches by targeting different proteins and inhibiting kinases [100]. It has been revealed in previous research that the CDK inhibitor Dinaciclib can bind to the KAc-binding pocket of BRDT, offering initial evidence for the potential repurposing of kinase inhibitors in the development of next-generation BRD4 inhibitors [101]. This discovery has inspired the development of a series of dual BET/kinase inhibitors [102,103]. Histone deacetylases (HDACs) reduce lysine acetylation, modulating BET protein binding to gene loci and influencing transcription [104,105]. Therefore, they are considered a rational strategy to suppress oncogene transcription and inhibit tumor cell proliferation more effectively [106,107]. A number of dual BET/HDAC inhibitors (59–64, Fig. 6) have demonstrated promising efficacy in the treatment of colorectal cancer (CRC) and AML, effectively suppressing the growth of BET inhibitor–resistant cells. These findings highlight the rising importance of dual-target drug development focusing on both BET proteins and HDACs [107–111].

For instance, compound 59 was designed and synthesized by combining (+)-JQ1 with HDAC inhibitors SAHA and CI-944 [112]. Structural modifications of already developed dual inhibitors can yield new derivatives with distinct biological activities. For example, compounds 66, 67, and 68 were optimized from the parent compound 65 [111,113] Other recently developed dual BET/HDAC inhibitors (69–73, Fig. 6) have followed similar design strategies, providing a broad reference framework for the study and development of this class of inhibitors [114–116].

The inhibition of BRD4, in conjunction with poly(ADP-ribose) polymerase 1 (PARP1) inhibitors, has been shown to induce homologous recombination (HR) deficiency, offering a promising new approach to cancer treatment [117,118]. Leveraging synthetic lethality, compound 74 (Fig. 6) represents the first dual BRD4/PARP1 inhibitor reported to induce apoptosis and G1-phase arrest in breast cancer cells effectively [119]. Huang et al. developed a dual-target PARP1/BRD4 inhibitor, III-7 (75), by integrating the PARP1 inhibitor Veliparib with a BRD4 inhibitor, showing efficacy in pancreatic cancer treatment [120]. Zhang et al. reported the identification of BP44 (76), a 4-hydroxyquinazoline derivative, as a dual BET/PARP1 inhibitor demonstrating strong antiproliferative activity in TNBC models [121]. It was revealed in another study that Polo-like kinase 1 (PLK1) forms a feed-forward loop with MYC through the upregulation of BRD4-controlled MYC transcription and protein stability [122]. Through chemical modification of the known BRD4/PLK1 dual inhibitor BI-2536 (65), WNY0824 (66), a new dual BET/PLK1 inhibitor created by modifying BI-2536 (65), effectively blocks BRD4-dependent androgen receptor signaling and MYC expression. Compounds 67 and 68 were also identified using this structure-guided approach [113,123]. However, a critical design challenge remains in achieving balanced selectivity between BET proteins and specific kinases; for example, some studies suggest that PLK1 inhibition, rather than BRD4, may be the primary driver of BI-2536's anticancer effects in MV4–11 cells [124].

Dual-target inhibitors with BET inhibitory activity have been serendipitously discovered. TG101209 and BI-2536, initially characterized as Polo-like kinase 1 (PLK1) and Janus kinase 2 (JAK2) inhibitors, respectively, were subsequently found to be strong BRD4 inhibitors [125]. Other kinase targets to consider include PI3K-AKT-mTOR pathway inhibitors, which control MYC expression, and CDK9 inhibitors, which enhance transcriptional elongation by phosphorylating Ser2 on RNA polymerase II's CTD [7,32,126]. Additionally, Casein Kinase 2 (CK2) aids in chromatin targeting and cofactor recruitment by phosphorylating BRD4. Gene regulation is collaboratively controlled by both Enhancer of Zeste Homolog 2 (EZH2) and BRD4, making them promising targets for developing dual-inhibitors in cancer therapy treatment [33,127]. A pharmacophore fusion strategy

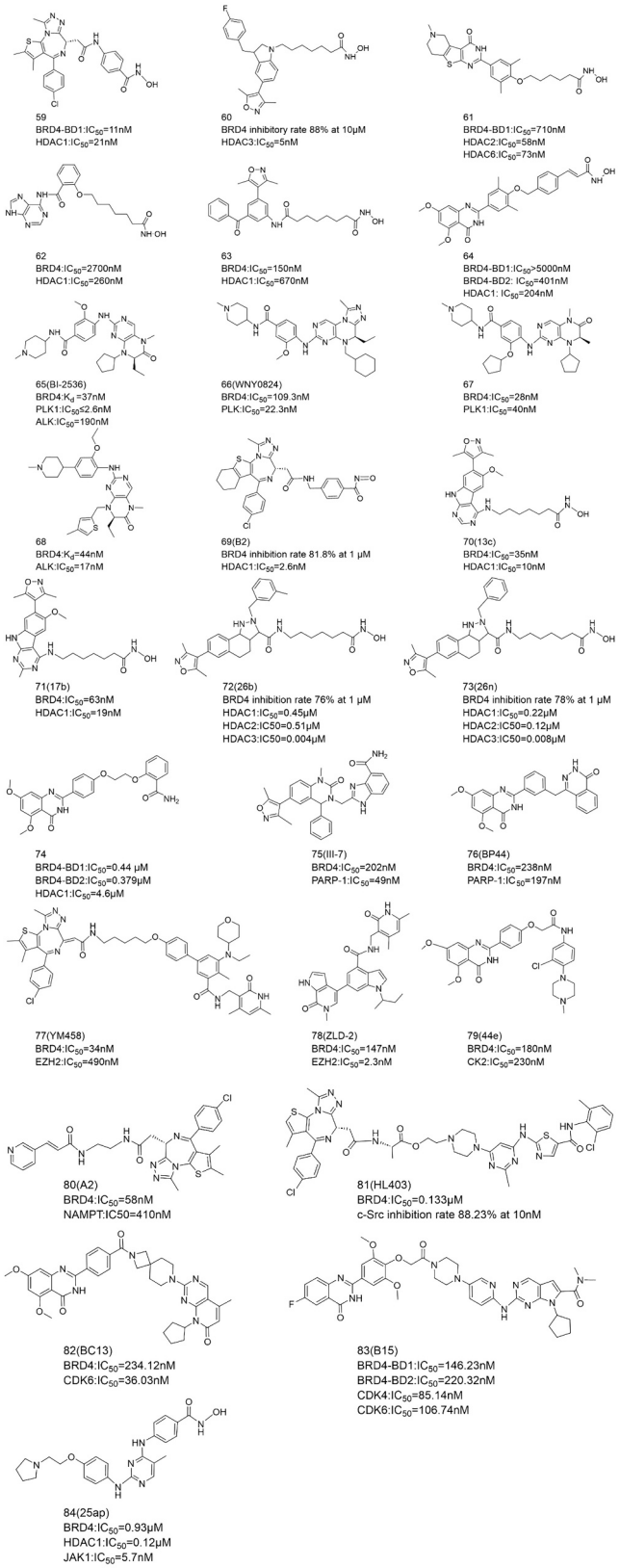


Fig. 6. Chemical structures of BRD4 dual-target/multi-target inhibitors 59–84.

based on EPZ6438 (EZH2 inhibitor) and JQ1 (BRD4 inhibitor) led to the design of YM458 (77), which significantly inhibited the proliferation of various solid tumor cell lines [127]. Likewise, compound ZLD-2 (78), derived from the fusion of GSK126 (EZH2 inhibitor) and JQ1, demonstrated dual inhibitory activity against EZH2 and BRD4 [128].

A range of dual and multi-target inhibitors — including BRD4/CK2 dual inhibitor 44e (79), BRD4/NAMPT (Nicotinamide Phosphoribosyltransferase) dual inhibitor A2 (80), BRD4/Src dual inhibitor HL403 (81), BRD4/CDK6 (Cyclin-Dependent Kinase 6) dual inhibitor BC13 (82), BRD4/CDK4 (Cyclin-Dependent Kinase 4)/6 dual inhibitor B15 (83), and the Fedratinib-based HDAC/JAK (Janus Kinase)/BRD4 triple inhibitor 25ap (84) — have shown promising therapeutic efficacy across multiple cancer models [129–134].

4.6. PROTACs

PROTACs (Proteolysis Targeting Chimeras) are an emerging strategy for functional protein suppression that operates through compound-mediated proteolysis. This approach exhibits unique pharmacological characteristics, including potent antiproliferative effects and reduced toxicity [1,19,22,135]. Most of the BET degraders developed to date function by harnessing ligands that engage the E3 ubiquitin ligase components Cereblon (CRBN) or Von Hippel–Lindau (VHL). ARV-771 (85, Fig. 7) is a pan-BET degrader that recruits CRBN and has been shown to downregulate androgen receptor (AR) signaling proteins in a CRPC xenograft model. It represents the first BET degrader reported for solid tumors [136]. MZ1 (86), a JQ1-based BRD4 degrader, was further optimized into the macrocyclic PROTAC compound macroPROTAC-1 (87) by incorporating a cyclized linker into the ARV-771 scaffold, resulting in comparable cellular potency [137,138]. dBET1 (88), another selective BET degrader, exhibits stronger inhibition than JQ1 in various cell lines, including MV4–11 [139]. Several additional JQ1-based PROTACs, such as AGB1 (89), MZP-54 (90), MZP-55 (91), and MZP-61 (92), have demonstrated high efficacy in inducing BET protein degradation [140,141]. Non-selective BET inhibitors combined with CRBN ligands have also been used to develop BRD4-selective degraders such as BD-7148 (93) and BD-9136 (94), which achieve high specificity for BRD4 [142].

Other BET degraders, including ARV-825 (95), BETd-246 (96), BETd-260 (97), DP1 (98), QCA570 (99), and L134 (100), have shown potent efficacy in models of lymphoblasts, xenografted lymphomas, and breast cancer [19,79,143–145]. Notably, ZXH-3-26 (101) and AT1 (102) are characterized by their negligible activity against BRD2 and BRD3 at optimal concentrations, enhancing their BRD4 selectivity [146,147].

Dual-target BRD4 PROTACs offer promising new directions in drug discovery. For instance, BRD4-PROTAC WWL0245 (103), developed from the dual BET/PLK1 inhibitor WNY0824, displays strong anticancer activity in AR-positive prostate cancer cells. DP-15 (104), targeting both BRD4 and GSPT1, is effective against AML and non-Hodgkin lymphoma models [148,149].

While demonstrating significant promise through low-dose, durable target inhibition, the clinical translation of PROTACs faces challenges. These include reliance on the ubiquitin-proteasome system (UPS), potential for efflux by transporters like ABCB1, adaptive resistance mechanisms such as WNT-MYC pathway bypass, and the hook effect limiting the therapeutic window. To overcome these hurdles, active research is exploring innovative strategies, including the development of molecular glues, nano-responsive delivery systems, and optimized catalytic warheads/ligands [150–152]. Antibody–PROTAC conjugates (Ab-PROTACs) offer a promising strategy for tissue-specific BRD4 degradation. For example, a trastuzumab-PROTAC conjugate can release the active PROTAC selectively upon linker cleavage, providing proof-of-concept for targeted degradation [153]. Selective degradation of BET proteins involves multiple pathways, including the ubiquitin–proteasome system as well as lysosomal and autophagy mechanisms [154].

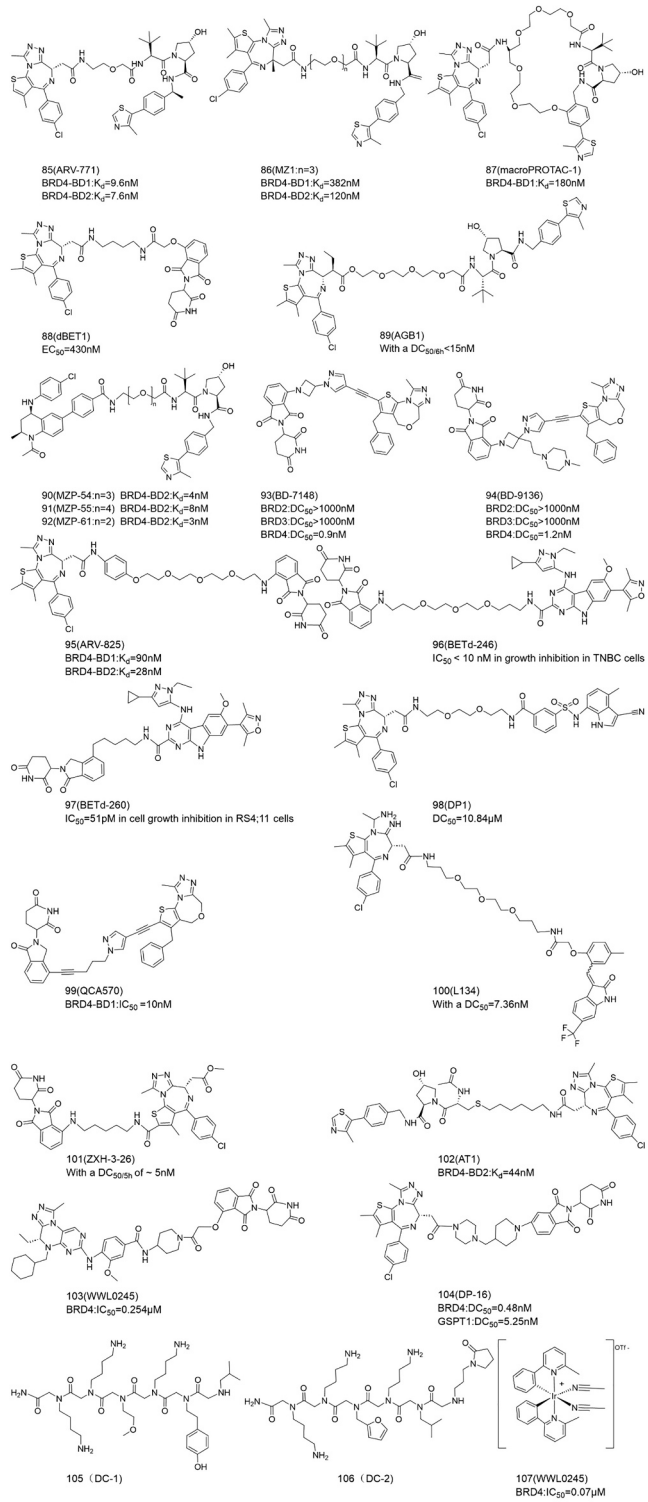


Fig. 7. Chemical structures of PROTAC degraders and PPI inhibitors 85–107.

4.7. Protein–protein interaction networks

Protein–protein interactions (PPIs) serve as crucial signaling junctions within molecular networks of normal and cancer cells, mediating the flow of pathophysiological information. Targeting these interactions to disrupt PPIs has thus emerged as a promising strategy to impede network signaling [155]. From a BRD38500-derived peptide combinatorial library, two peptide-like ligands—DC-1 (105) and DC-2 (106)

(Fig. 7)—were identified that bind selectively to a phosphorylated β motif within human BRD4 [156]. These discoveries offer novel templates for PPI-targeted BRD4 inhibitor development [35,157]. Additionally, the irreversible PPI inhibitor 48 (107) has been shown to block the interaction between BRD4-BD1 and tetra-acetylated histone H4 peptides, thereby suppressing BRD4-driven transcription of MYC and BCL-2 genes [158]. Despite these promising leads, PPI-directed targeting of BRD4 remains underexplored. The main reasons include difficulties in confirming in vivo targets and action mechanisms, as well as obstacles in setting up effective high-throughput screening systems for PPI inhibitors [159].

5. Conclusion and future perspectives

To conclude, the development of BRD4 inhibitors has progressed significantly, with many institutions filing patents that reflect strong engagement in this field. Selective inhibitors targeting BRD4-BD1/BD2 and dual-target BRD4 inhibitors have gained prominence over pan-BET inhibitors as key research priorities. Both clinical and in vitro data demonstrate that selective inhibitors disrupt only disease-induced aberrant transcription with minimal impact on basal gene expression, thereby improving the therapeutic index and offering a more tolerable strategy for long-term treatment of non-oncologic conditions such as cardiovascular and metabolic diseases [160]. The development of BRD4-selective inhibitors, however, remains in its infancy [161,162]. The design of more selective inhibitors is becoming increasingly feasible due to enhanced understanding of the structural biology of BET proteins. Advances in protein structure prediction technologies now allow the simulation and refinement of protein conformations to better approximate native structures. These tools can be used to explore whether and how small molecules interact with BRD4 domains, providing new strategies for the rational design of selective inhibitors [163,164]. PROTAC and PPI technologies offer promising alternatives for targeting BRD4. PROTACs enable potent, sustained degradation at low doses, although their clinical translation presents significant challenges. Realizing the full potential of these innovative approaches requires overcoming these hurdles through continued collaborative efforts.

CRediT authorship contribution statement

Linlin Deng: Writing – review & editing, Writing – original draft, Methodology. **Erkang Tian:** Supervision, Formal analysis, Conceptualization. **Ting Ma:** Formal analysis, Data curation. **Shaojie Liang:** Validation, Investigation, Conceptualization. **Sijia Hu:** Methodology. **Linwei Li:** Writing – original draft, Methodology. **Juan Li:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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